can be achieved. All the Chebyshev filters designed by using Table 3-1 have a passband ripple of about 6% (0.5 dB), a good compromise, and a common choice. A similar design, the **elliptic filter**, allows ripple in both the passband *and* the stopband. Although harder to design, elliptic filters can achieve an even better tradeoff between roll-off and passband ripple.

In comparison, the Butterworth filter is optimized to provide the sharpest roll-off possible *without* allowing ripple in the passband. It is commonly called the *maximally flat filter*, and is identical to a Chebyshev designed for zero passband ripple. The Bessel filter has no ripple in the passband, but the roll-off far worse than the Butterworth.

The last parameter to evaluate is the **step response**, how the filter responds when the input rapidly changes from one value to another. Figure 3-13 shows the step response of each of the three filters. The horizontal axis is shown for filters with a 1 hertz cutoff frequency, but can be scaled (inversely) for higher cutoff frequencies. For example, a 1000 hertz cutoff frequency would show a step response in *milliseconds*, rather than *seconds*. The Butterworth and Chebyshev filters **overshoot** and show **ringing** (oscillations that slowly decreasing in amplitude). In comparison, the Bessel filter has neither of these nasty problems.

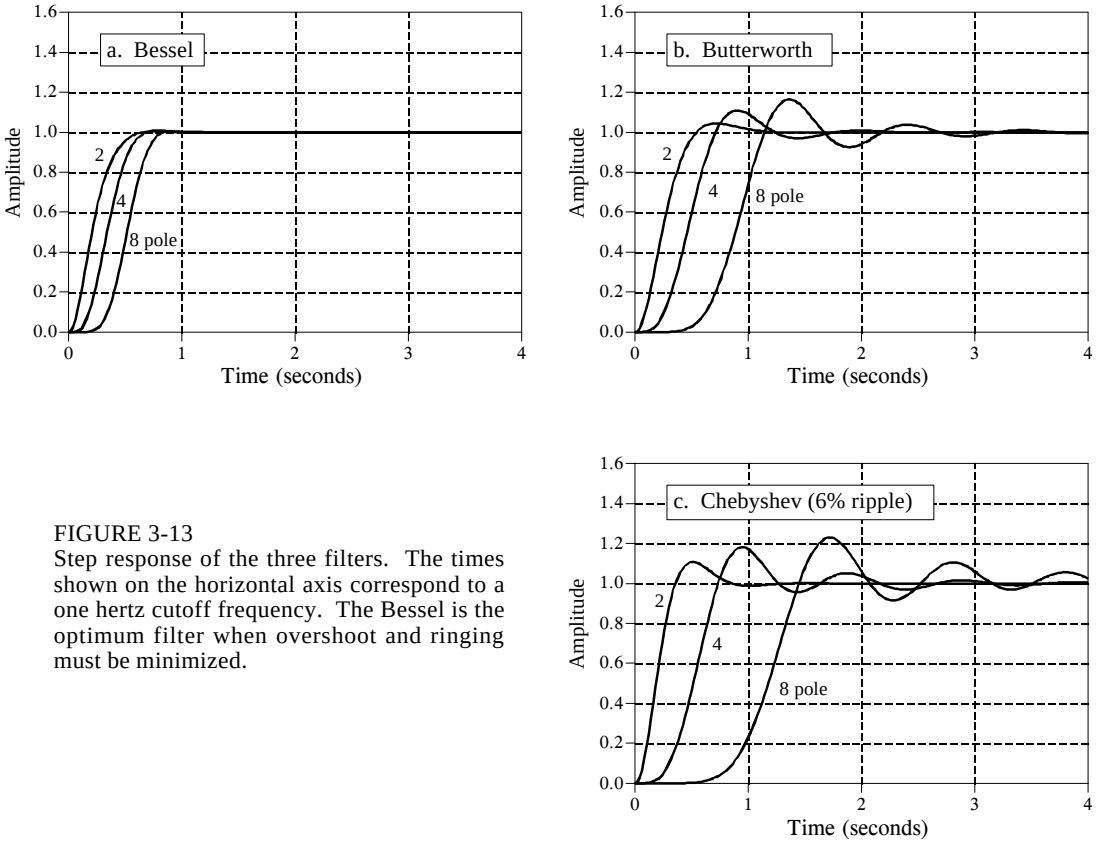

FIGURE 3-13
Step response of the three filters. The times shown on the horizontal axis correspond to a one hertz cutoff frequency. The Bessel is the optimum filter when overshoot and ringing must be minimized.